

SAMBPS Protection Framework

Adaptive relay scheme for inverter-rich distribution networks

Tier 1 — Local Relay Algorithms

Ch. 4–6

87L Differential Suite (Ch. 4)

- IBR-adaptive slope
- Current-differential
- Phase-segregated logic

87T SPRT (Ch. 5)

- Sequential prob. ratio
- Inrush discrimination
- Adaptive k_m threshold

Generator Suite (Ch. 6)

- 40, 64G, 81 elements
- IBR mode-detection
- Confidence-gate veto

$\hat{k}_{ibr}, \hat{\alpha}, \kappa_n$



Tier 2 — System Integration

Ch. 7

5-Layer Architecture (C16)

- Digital twin EKF
- k_{ibr} / α est.
- Confidence-gate veto

Coordination & HIL (C19–20)

- N-1 CTI protocol
- HIL: 62 scenarios
- GOOSE security

Classical protection framework

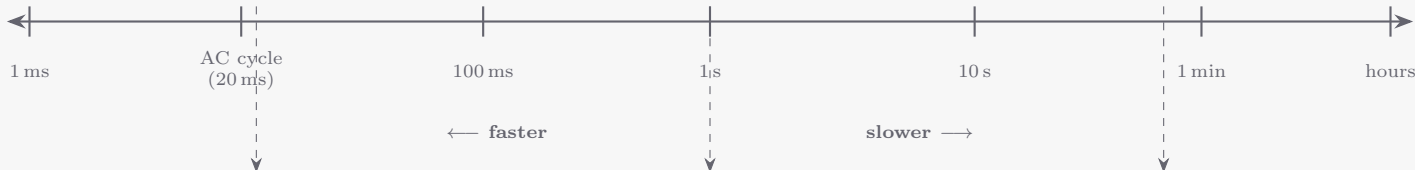
Overcurrent / fuses
sub-cycle to 1 s

Distance / differential
cycles to seconds

Protection coordination
seconds to minutes

Tier 1 (1 ms – 200 ms)

Tier 2 (100 ms – min)



IEC 61850 GOOSE + IEEE C37.118 PMU Network